

TE Curve Verification Pipeline Familywise ONNX Report - wave4_4_gru_latent_offset_residual

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave4_4_gru_latent_offset_residual/2026-07-15-22-18-25__track2_wave4_4_gru_latent_offset_residual_familywise_onnx_report ;`
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave4_4_gru_latent_offset_residual/2026-07-15-22-18-25__track2_wave4_4_gru_latent_offset_residual_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml ;`
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_4_gru_latent_offset_residual/2026-07-15-22-18-25__track2_wave4_4_gru_latent_offset_residual_familywise_onnx_report/model_inventory.csv ;`
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_4_gru_latent_offset_residual/2026-07-15-22-18-25__track2_wave4_4_gru_latent_offset_residual_familywise_onnx_report/per_curve_metrics.csv .`

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave4_4_gru_latent_offset_residual` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave4_4_gru_latent_offset_residual_fw_simplified_setpoints</code>	<code>2026-07-15-07-11-49_te_wave4_4_gru_latent_offset_residual_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_wave4_4_gru_latent_offset_residual_bw_simplified_setpoints</code>	<code>2026-07-15-07-26-38_te_wave4_4_gru_latent_offset_residual_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_wave4_4_gru_latent_offset_residual_global_simplified_setpoints</code>	<code>2026-07-15-06-59-43_te_wave4_4_gru_latent_offset_residual_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/forward/2026-07-15-07-11-49_te_wave4_4_gru_latent_offset_residual_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/forward/2026-07-15-07-11-49_te_wave4_4_gru_latent_offset_residual_fw_simplified_setpoints/python/latent_state_hysteresis_probe-epoch=081-val_mae=0.00371922.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/backward/2026-07-15-07-26-38_te_wave4_4_gru_latent_offset_residual_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/backward/2026-07-15-07-26-38_te_wave4_4_gru_latent_offset_residual_bw_simplified_setpoints/python/latent_state_hysteresis_probe-epoch=072-val_mae=0.00377245.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/global/2026-07-15-06-59-43_te_wave4_4_gru_latent_offset_residual_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/global/2026-07-15-06-59-43_te_wave4_4_gru_latent_offset_residual_global_simplified_setpoints/python/latent_state_hysteresis_probe-epoch=065-val_mae=0.00375664.ckpt</code>

Aggregate Metrics

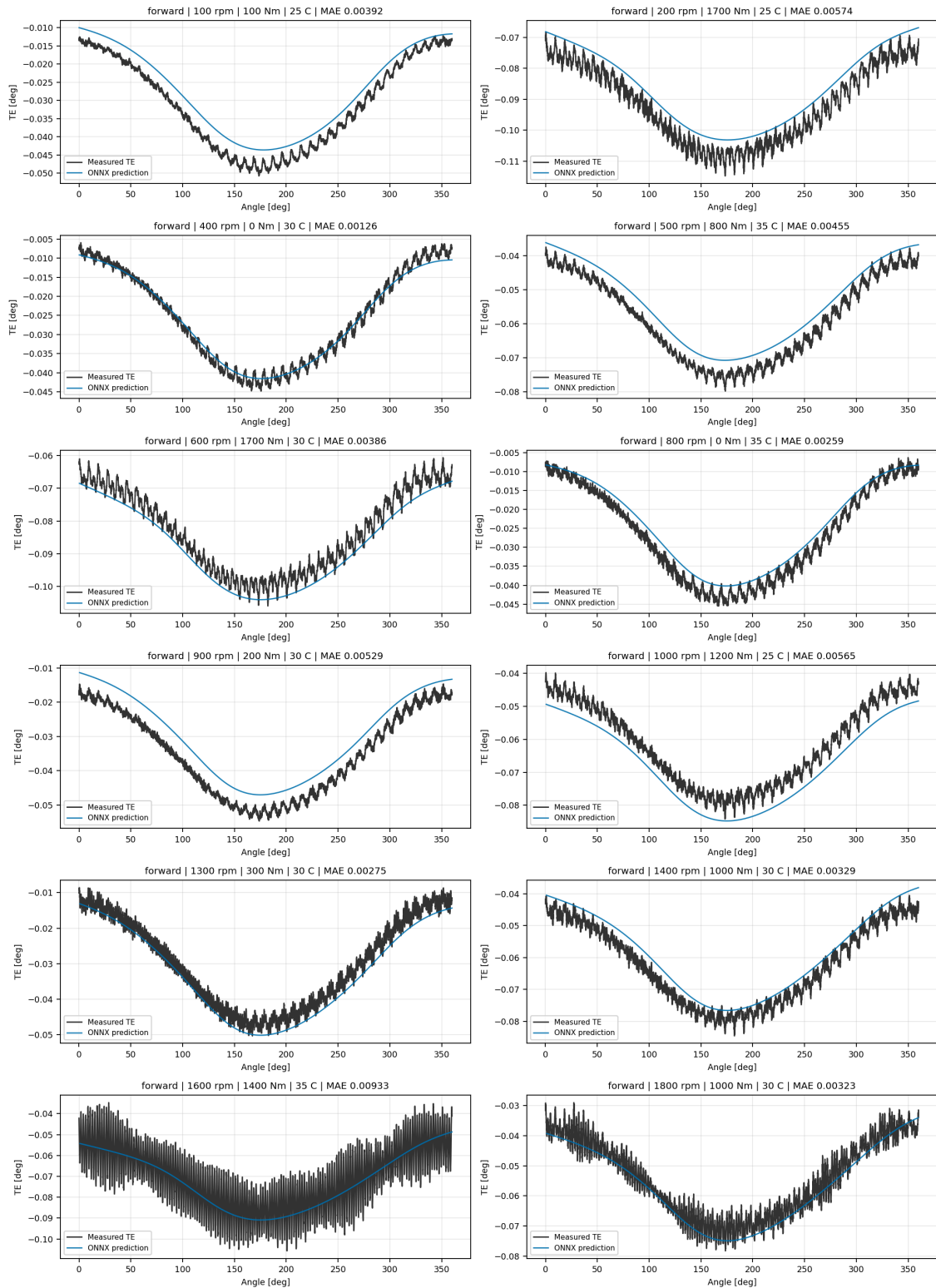
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003522	0.003976	8.251	13.855
backward	97	0.003631	0.004129	8.429	13.607
global	194	0.003535	0.003979	8.242	13.654

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000147	0.002951	0.001819	0.005940
backward	-0.000013	0.002861	0.002036	0.005469
global	-0.000337	0.002912	0.001827	0.004557

Forward 12-Curve Page

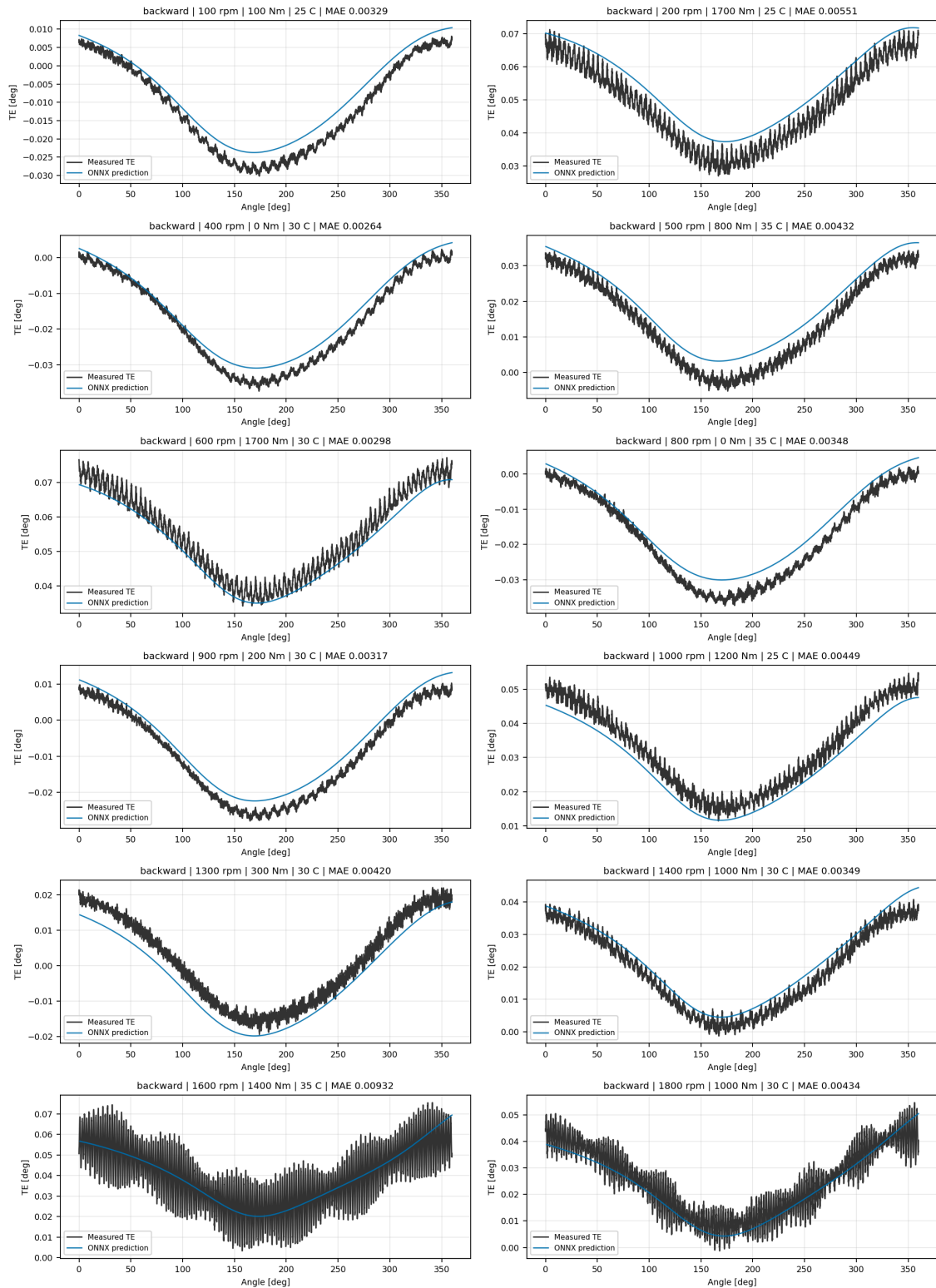
Simplified Dataset + Setpoints | wave4_4_gru_latent_offset_residual | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

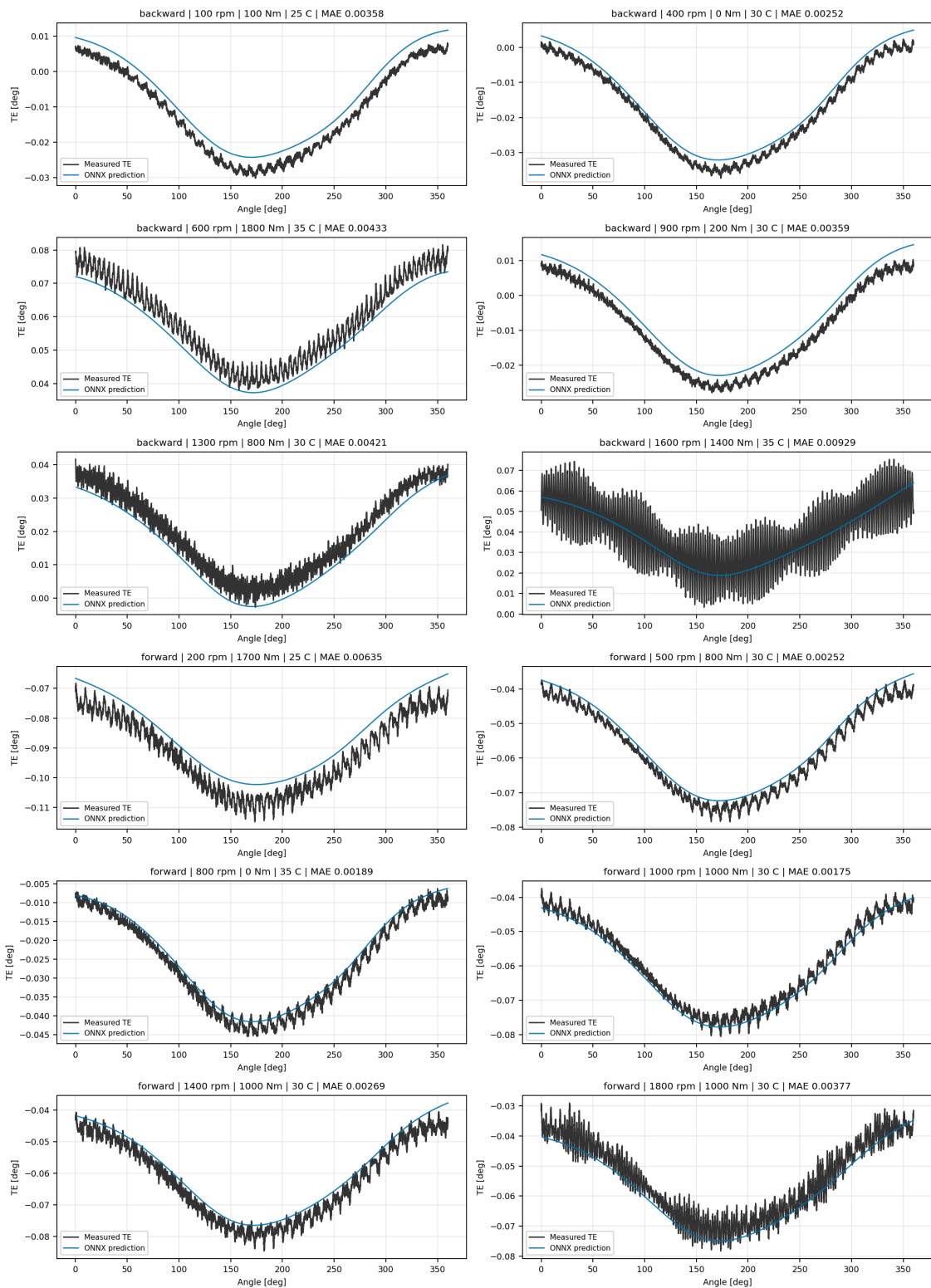
Simplified Dataset + Setpoints | wave4_4_gru_latent_offset_residual | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave4_4_gru_latent_offset_residual | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave4_4_gru_latent_offset_residual ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_4_gru_latent_offset_residual_fw_polished_setpoints	2026-07-15-17-03-49__te_wave4_4_gru_latent_offset_residual_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_wave4_4_gru_latent_offset_residual_bw_polished_setpoints	2026-07-15-17-34-55__te_wave4_4_gru_latent_offset_residual_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_wave4_4_gru_latent_offset_residual_global_polished_setpoints	2026-07-15-16-39-39__te_wave4_4_gru_latent_offset_residual_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/forward/2026-07-15-17-03-49__te_wave4_4_gru_latent_offset_residual_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/forward/2026-07-15-17-03-49__te_wave4_4_gru_latent_offset_residual_fw_polished_setpoints/python/latent_state_hysteresis_probe-epoch=164-val_mae=0.00221821.ckpt
backward	models/polished_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/backward/2026-07-15-17-34-55__te_wave4_4_gru_latent_offset_residual_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/backward/2026-07-15-17-34-55__te_wave4_4_gru_latent_offset_residual_bw_polished_setpoints/python/latent_state_hysteresis_probe-epoch=036-val_mae=0.00226526.ckpt
global	models/polished_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/global/2026-07-15-16-39-39__te_wave4_4_gru_latent_offset_residual_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_4_gru_latent_offset_residual/global/2026-07-15-16-39-39__te_wave4_4_gru_latent_offset_residual_global_polished_setpoints/python/latent_state_hysteresis_probe-epoch=100-val_mae=0.00222289.ckpt

Aggregate Metrics

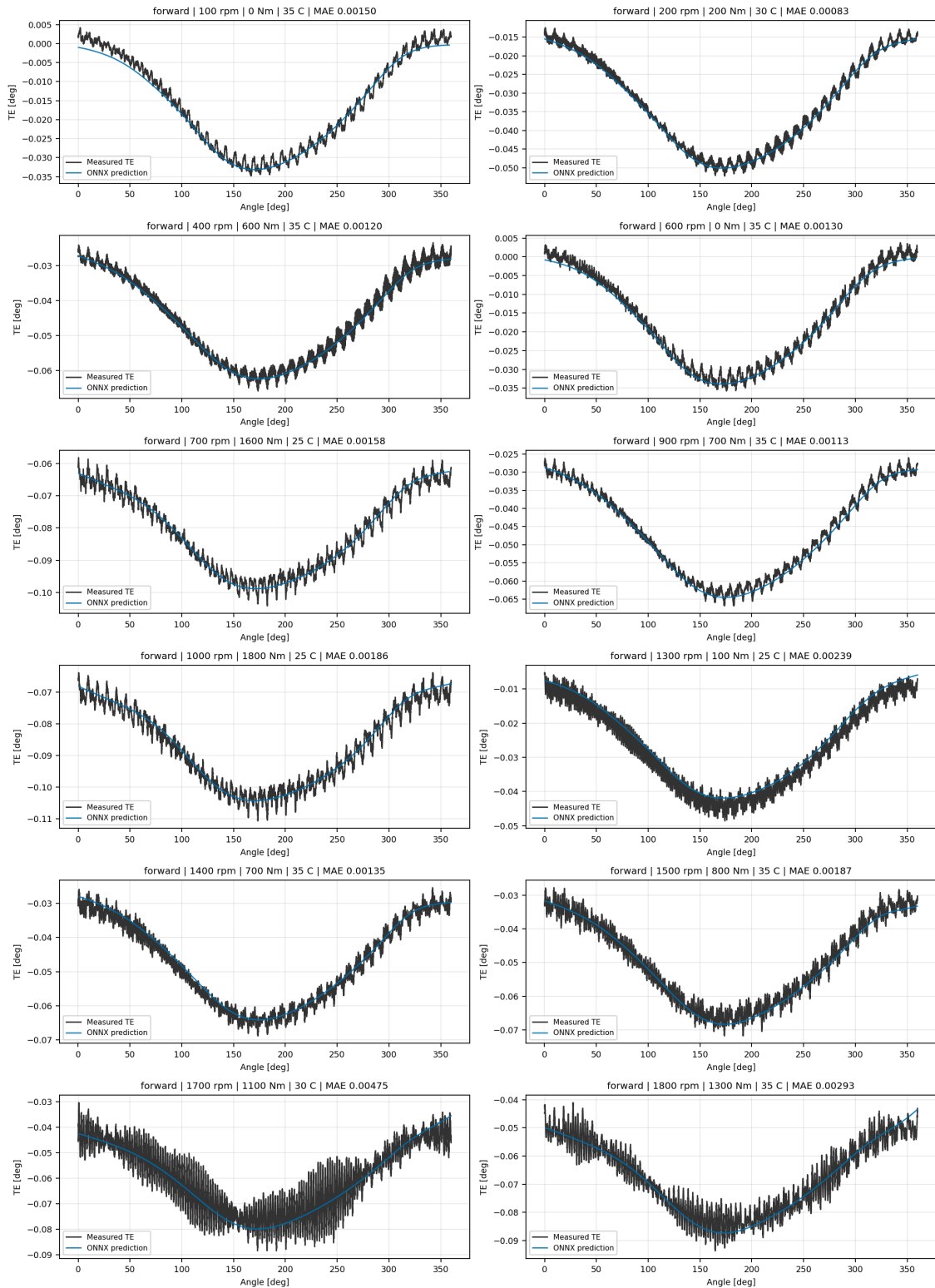
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002266	0.002721	5.038	10.929
backward	94	0.002836	0.003378	5.522	11.514
global	194	0.002534	0.003041	5.241	11.510

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000264	0.000923	0.001948	0.005463
backward	0.000055	0.001090	0.002443	0.006519
global	-0.000057	0.001054	0.002169	0.006334

Forward 12-Curve Page

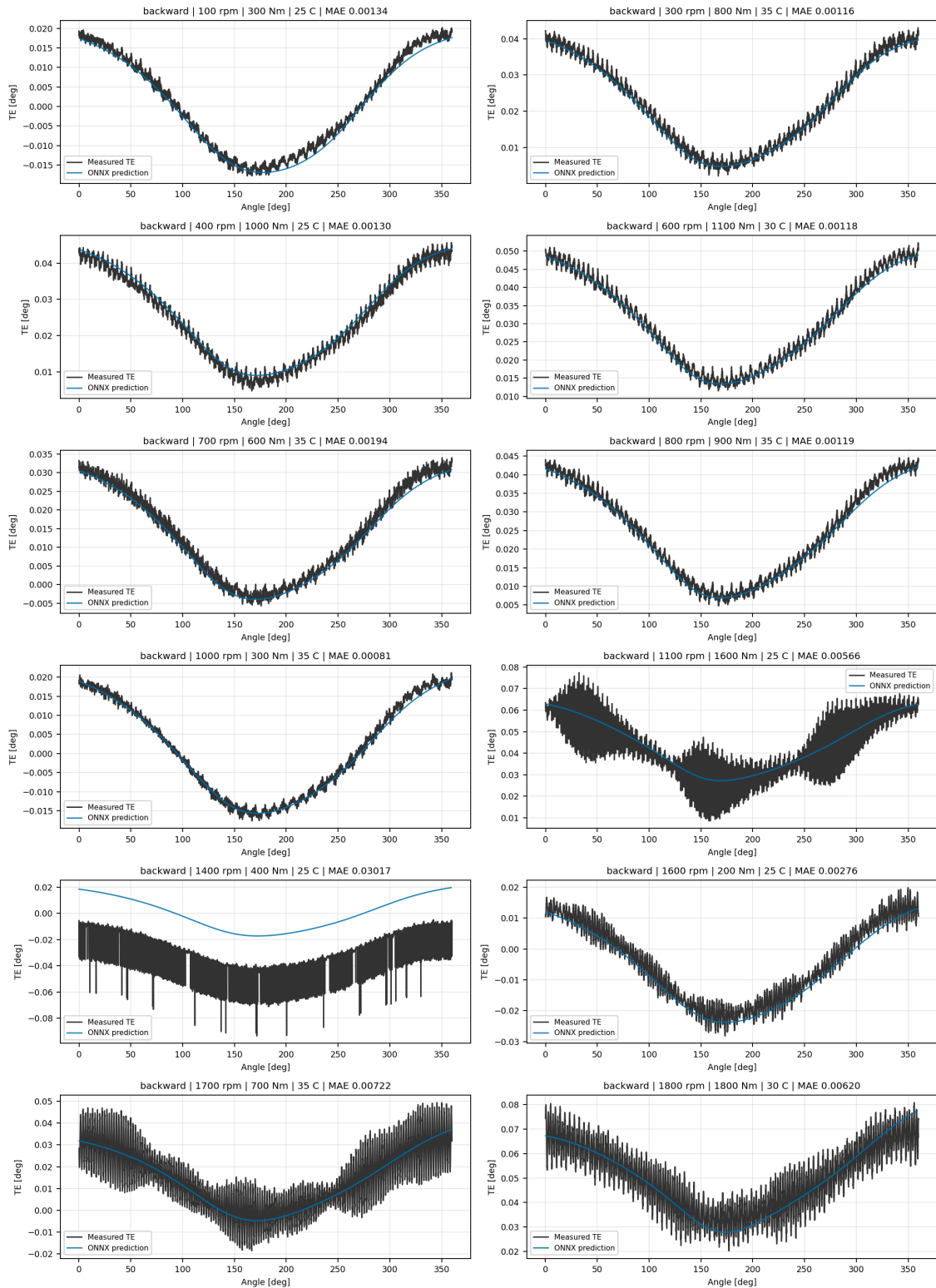
Polished Dataset + Setpoints | wave4_4_gru_latent_offset_residual | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

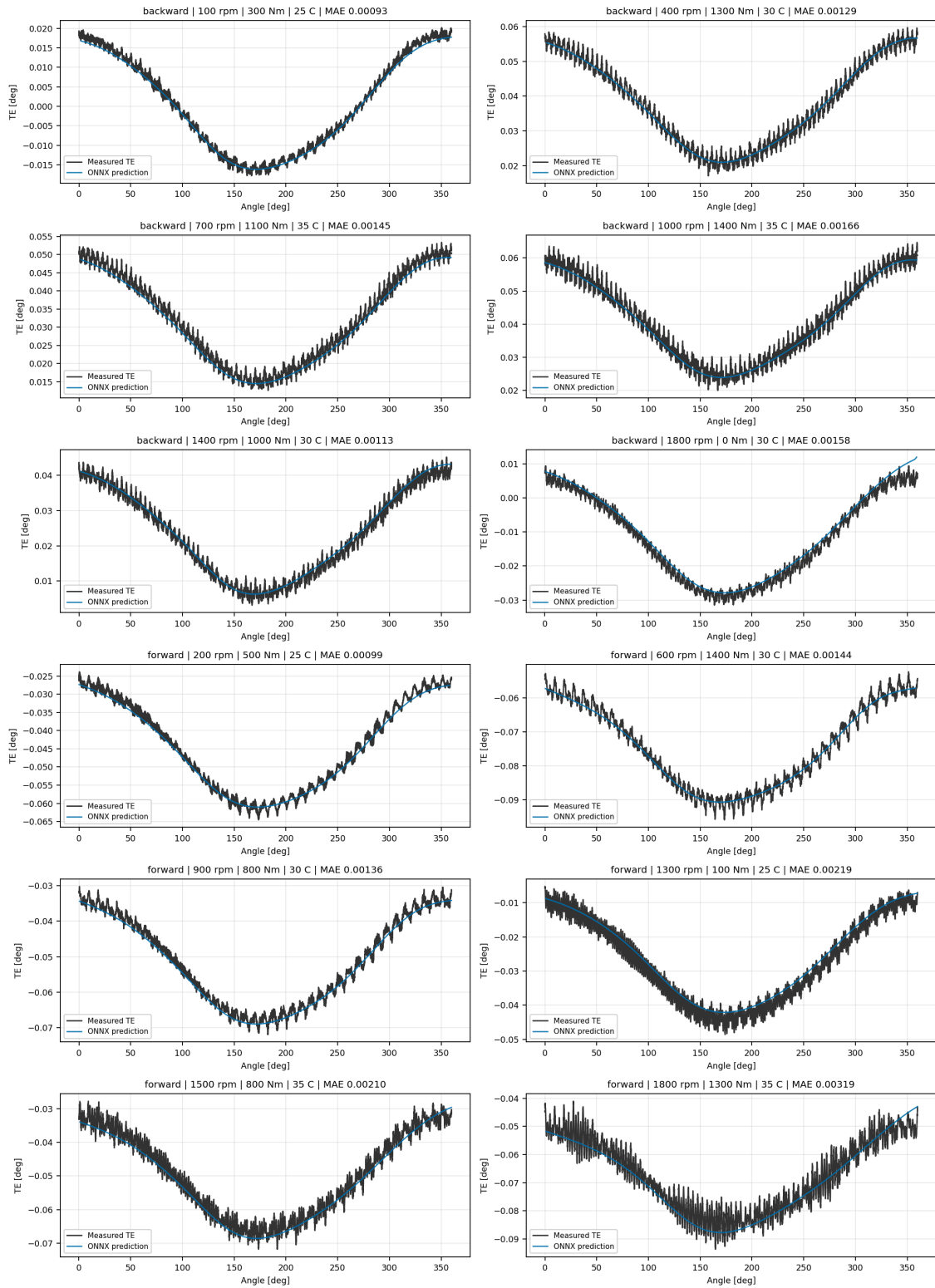
Polished Dataset + Setpoints | wave4_4_gru_latent_offset_residual | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave4_4_gru_latent_offset_residual | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave4_4_gru_latent_offset_residual ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_4_gru_latent_offset_residual_fw_polished_actual_values	2026-07-15-18-51-23__te_wave4_4_gru_latent_offset_residual_fw_polished_actual_values	polished_point_v1
backward	te_wave4_4_gru_latent_offset_residual_bw_polished_actual_values	2026-07-15-19-13-13__te_wave4_4_gru_latent_offset_residual_bw_polished_actual_values	polished_point_v1
global	te_wave4_4_gru_latent_offset_residual_global_polished_actual_values	2026-07-15-18-05-18__te_wave4_4_gru_latent_offset_residual_global_polished_actual_value s	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave4_4_gru_latent_offset_residual/forward/2026-07-15-18-51-23__te_wave4_4_gru_latent_offset_residual_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_4_gru_latent_offset_residual/forward/2026-07-15-18-51-23__te_wave4_4_gru_latent_offset_residual_fw_polished_actual_values/python/latent_state_hysteresis_probe-epoch=113-val_mae=0.00224746.ckpt
backward	models/polished_dataset/actual_values/exported/wave4_4_gru_latent_offset_residual/backward/2026-07-15-19-13-13__te_wave4_4_gru_latent_offset_residual_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_4_gru_latent_offset_residual/backward/2026-07-15-19-13-13__te_wave4_4_gru_latent_offset_residual_bw_polished_actual_values/python/latent_state_hysteresis_probe-epoch=140-val_mae=0.00222826.ckpt
global	models/polished_dataset/actual_values/exported/wave4_4_gru_latent_offset_residual/global/2026-07-15-18-05-18__te_wave4_4_gru_latent_offset_residual_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_4_gru_latent_offset_residual/global/2026-07-15-18-05-18__te_wave4_4_gru_latent_offset_residual_global_polished_actual_values/python/latent_state_hysteresis_probe-epoch=208-val_mae=0.00217281.ckpt

Aggregate Metrics

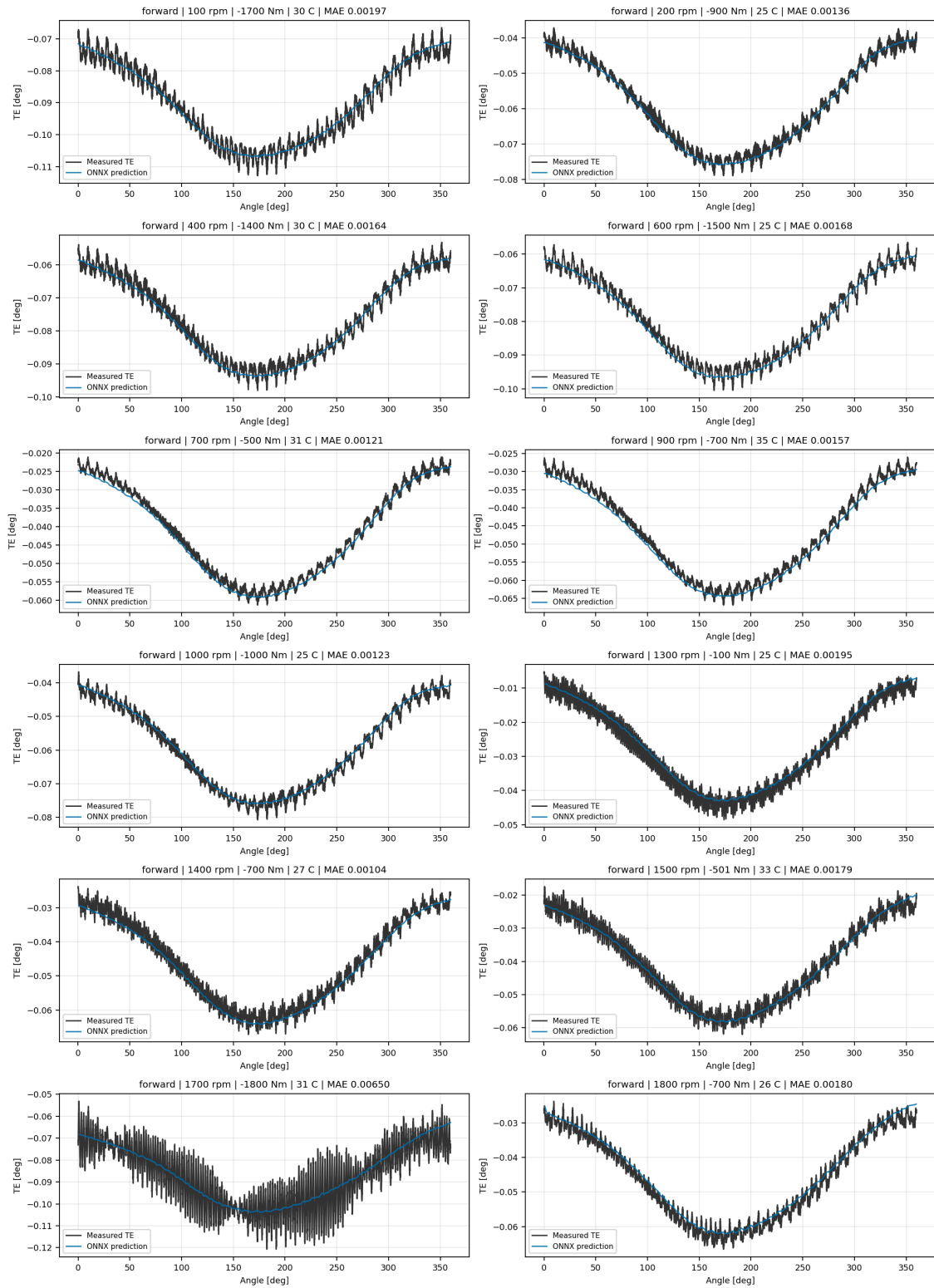
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002210	0.002681	4.885	9.768
backward	94	0.002547	0.003098	5.046	11.669
global	194	0.002271	0.002760	4.701	11.037

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000505	0.000889	0.001982	0.005484
backward	-0.000211	0.000681	0.002406	0.005017
global	-0.000072	0.000628	0.002142	0.004824

Forward 12-Curve Page

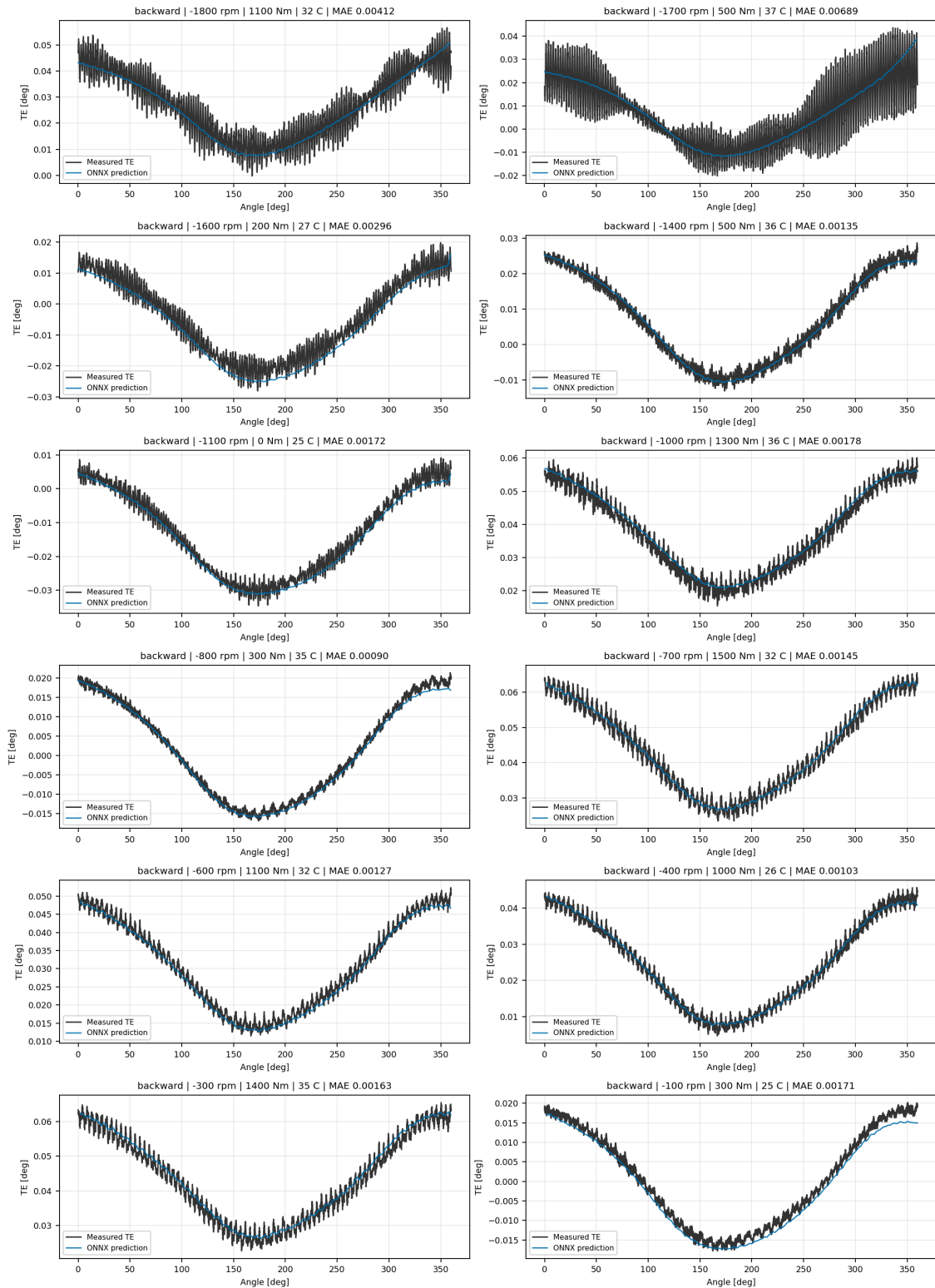
Polished Dataset + Actual Values | wave4_4_gru_latent_offset_residual | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

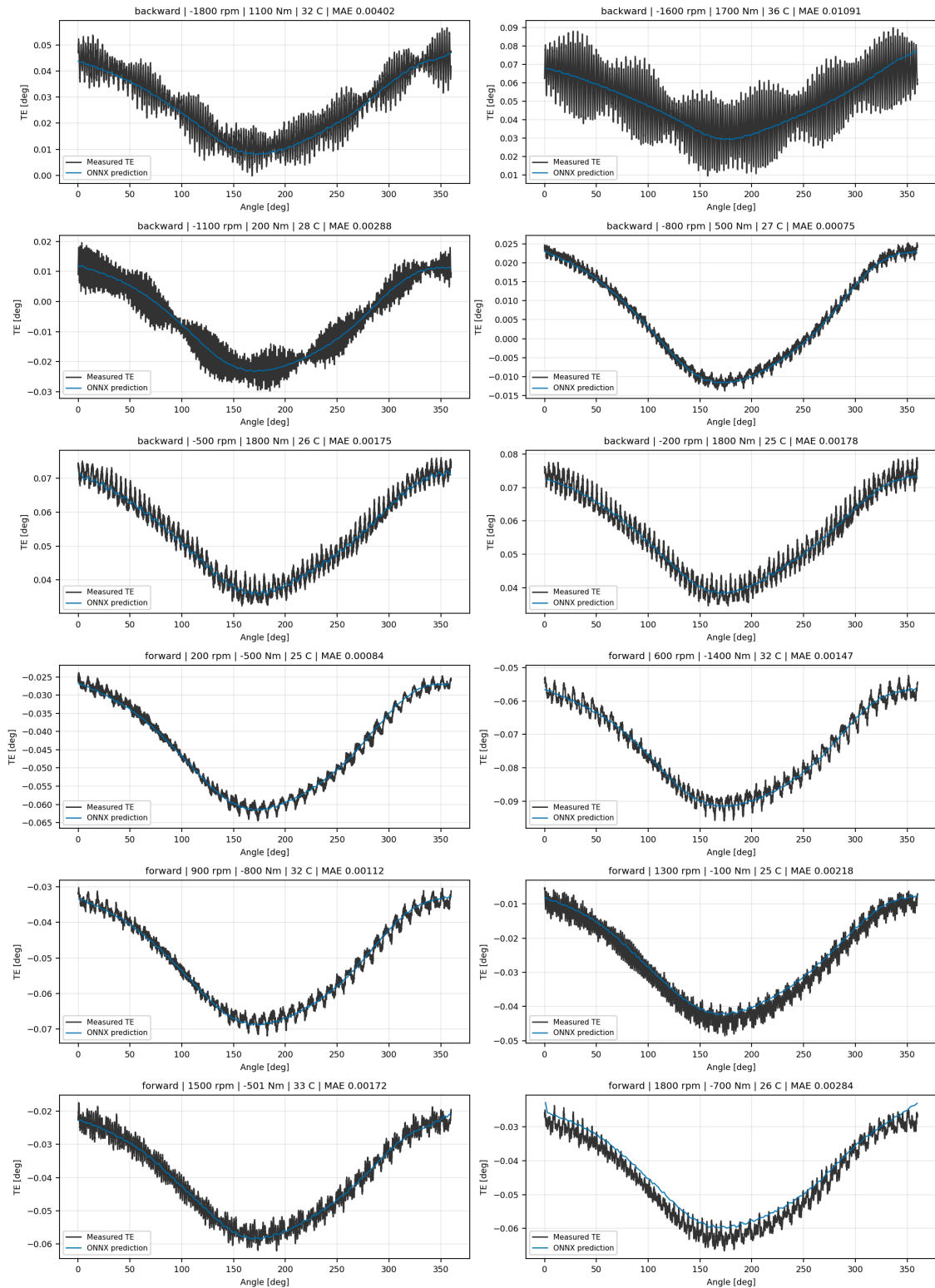
Polished Dataset + Actual Values | wave4_4_gru_latent_offset_residual | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave4_4_gru_latent_offset_residual | global



polished_dataset_actual_values global 12-curve collage